

LINGAM THUVARAKAN

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PROFESSIONAL PROFILE

Motivated and detail-oriented Computer Science undergraduate (CGPA: 3.7/4.0) with strong expertise in full-stack development (Next.js, TypeScript), mobile application engineering (Flutter), secure systems, and blockchain architectures. Experienced in building scalable applications using MongoDB/MySQL and delivering reliable systems through testing, documentation, and project coordination. Strong leadership background with active volunteering and continuous skill development.

EDUCATION

Bachelor of Science (Hons) in Computer Science Jun 2022 – Present
University of Jaffna — CGPA: 3.7/4.0 (73 Credits)

G.C.E. Advanced Level (Mathematics Stream) 2020
Jaffna Hindu College, Jaffna — Results: 1A, 2B — Z-score: 1.301

PROJECTS

- 1. Blockchain Based Secure Election System (Academic Research)** Nov 2025 – Present
 - Designed a secure multi-district electronic voting architecture using a permissioned blockchain with PBFT consensus.
 - Implemented Tamper-Evidence-Based Consequence Validation (TECV) to detect manipulation and protect election outcomes.
 - Integrated biometric voter authentication and strong identity validation to ensure voter fairness.
 - Built append-only encrypted vote storage using CouchDB for tamper-evidence and audit support.
 - Engineered district-level blockchain ledgers to ensure integrity, immutability, and Byzantine fault tolerance.
- 2. Company Management System for Olir-Studios(Team Software Project)** Jan 2025 - Nov 2025
Role: Team Leader & Full-stack Developer
 - Designed and developed a full-stack system using Next.js and MongoDB with secure authentication via Clerk.
 - Built modern, responsive UI components using shadcn/ui and React Hooks with efficient state/component management.
 - Implemented WebSocket-based real-time chat functionality to support internal team communication.

- Developed and tested backend APIs, handled database integration, and performed end-to-end testing for reliability.
- Managed collaboration and task tracking using ClickUp to coordinate team delivery.

3. Efficient OGPA Calculator (Personal Project)

Dec 2025 – Jan 2026

- Developed a Flutter (Dart) mobile application for accurate Semester GPA and Overall GPA (OGPA) calculation using a credit-weighted formula.
- Implemented real-time GPA/OGPA recalculation, semester-wise tracking, and subject/credit management.
- Built automatic grade-to-point conversion and strong input validation to prevent incorrect academic entries.
- Designed a clean, modern, responsive UI and a visual academic performance summary dashboard.
- Added secure on-device local persistence for saving semesters and results.

4. My Digital Wallet for Individuals (Personal Project)

Dec 2025 – Feb 2026

- Designed and developed a comprehensive offline personal finance mobile application using Flutter (Dart) and Hive local storage.
- Implemented full CRUD for income/expense tracking with real-time analytics (daily, monthly, total summaries).
- Built 7-day cashflow visualization, savings pots with goal tracking, and progress indicators.
- Implemented bank account management with deposit/withdrawal features and integrated overall wealth calculation.
- Delivered a modular, scalable architecture with a modern Apple-inspired UI and planned enhancements (budgeting, category analytics, investments, backup/restore, biometric security, APK deployment).

5. Personal AI Desktop Assistant (Jarvis-Inspired System)

Feb 2026 – Present

- Developed a Windows voice assistant that runs as a background system service and starts automatically on boot.
- Integrated wake-word detection, real-time speech recognition, natural command parsing, and text-to-speech responses.
- Implemented secure command handling using allowlist-based controls and safe deletion via recycle-bin mechanisms.
- Enabled voice-driven system actions (open apps, manage folders, system-level operations) using a modular skill-based design.

6. Point of Sale (POS) System Development for JeyaFood (Team Project)

Jan 2026 – Present

- Developed a restaurant POS system using Next.js, TypeScript, and MongoDB supporting Dine-in and Cash-on-Delivery workflows.
- Implemented structured order status management, real-time order tracking, and secure transaction handling.
- Built role-based access control (Admin/Staff), receipt generation, and daily sales reporting for operational efficiency.
- Designed scalable order schemas with price snapshotting and audit logging to maintain transaction accuracy and integrity.

FUTURE PROJECTS IDEAS

1. Automated QA Testing Platform (VS Code Integrated)

Now Starts

- Designing an intelligent QA platform integrated with Visual Studio Code that detects forms/inputs and auto-generates validation test cases.
- Building a rule-based engine to infer constraints (required fields, data types, regex patterns) and support custom business rules.
- Implementing automated crawling from the root route and Playwright-based end-to-end testing with structured reporting.
- Goal: reduce manual QA effort, improve release reliability, and accelerate delivery cycles.

2. ServiceHub – Location-Based Multi-Vendor Services Marketplace

Now Starts

- Designing a location-based marketplace enabling customers to request services and receive structured quotations from multiple providers.
- Implementing job lifecycle management: Request → Quotes → Confirmed → In Progress → Completed → Paid.
- Adding secure Email OTP authentication, role-based access control, and privacy-first design (no direct contact exposure).
- Engineering sequential IDs (e.g., CUS00000001, PRO00000001) using atomic database counters and structured MongoDB schemas.
- Building an admin panel for provider approval, moderation, dispute handling, and commission tracking.

SKILLS & COMPETENCIES

Technical Skills: Manual Testing, Functional Testing, Regression Testing, Test Case Design, Test Planning, Defect Tracking, API Testing (Postman).

Tools: Postman, Git, GitHub, VS Code, ClickUp.

Programming: Java (OOP), Python, TypeScript, JavaScript, Dart.

Frameworks: Next.js, React, Flutter.

Databases: MongoDB, MySQL, CouchDB (Basic: Supabase).

Other: Basic understanding of Machine Learning, Statistics, and Version Control practices.

Soft Skills: Project Management, Teamwork, Time Management, Leadership, Effective Communication, Critical Thinking, Problem Solving, Attention to Detail, Analytical Thinking.

CERTIFICATIONS & E-LEARNING

Leadership Training Programme

2024

- Completed a structured leadership development programme focused on communication, teamwork, and decision-making.

Certificate Courses – Alison

- Java Programming for Complete Beginners
- Diploma in Digital Forensic Investigation

Certificate Courses – edX

- Python Basics for Data Science

Certificate Courses – University of Moratuwa

- Python Programming Level 1

VOLUNTEERING & LEADERSHIP

Science Union Secretary, Faculty of Science, University of Jaffna Jan 2025 – Jan 2026

- Coordinated faculty activities, student engagement programmes, and internal communications.
- Supported planning, execution, and documentation of Science Union events and responsibilities.

Leo Club of Nallur, Leo District 306 D12

Jun 2021 – Present

- Promote as Leo Lion Nov 2025 – Present
- IMM Past President & Editor Jun 2025 – Present
- Regional Chairperson Jun 2025 – Present
- Club President Jun 2024 – May 2025
- Club Vice President Jun 2023 – May 2024
- Club Treasurer Jun 2022 – May 2023
- Project Coordinator Jun 2021 – May 2022
- Completed 33 jobs and earned 3950 points in My Leo Portal (MD 306) Jun 2021 – Present

AWARDS

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| • Sri Lanka University Games (SLUG) – Volleyball (2nd Runners-up) | 2022 |
| • Top 10 Leos Awards – Leo District 306 B1 | 2022 |
| • Most Outstanding Vice-President – Leo District 306 B1 (2nd Runner-up) | 2023 |
| • Provincial Weightlifting Competition (Under-17 – Runner-up) | 2017 |

LANGUAGES

English (Fluent) — Tamil (Native)

REFEREES

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